Numerical techniques in Hydrodynamics.

Tutorial - 6

Lid - Driven cavity problem

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Incompressible fluid flow :-

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \rightarrow \text{conservation of mass.}$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu/\rho\left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right]$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial y} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu/\rho\left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right]$$

↳ momentum Equations For x and Y direction.

Vorticity - Stream Function

* Vorticity is measure the amount of Rotation in flow

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial v}$$

the fluid Rotation about the z axis.

$$\omega_z = \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right]$$

u, v values substitude in the continuity Equation,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial \left(\frac{\partial \varphi}{\partial y}\right)}{\partial x} + \frac{\partial \left(-\partial \psi / \partial x\right)}{\partial y} = 0$$

$\omega \rightarrow$ curl of velocity.

$$\frac{\partial \varphi}{\partial x \partial y} - \frac{\partial \psi}{\partial x \partial y} = 0$$

flow is conserved property.

the continuity is equal to zero.

$$\nabla \cdot \vec{V} = 0$$

For the vorticity (ω)

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}.$$

$$\omega = -\frac{\partial \left(\partial \psi / \partial x\right)}{\partial x} - \frac{\partial \left(\frac{\partial \psi}{\partial y}\right)}{\partial y}$$

$$\omega = \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial y^2}$$

$$\boxed{\omega = -\left[\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right]}$$

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega} \rightarrow \text{poisson's equation for stream function.}$$

For Eliminating the pressure term,

x momentum equation $\frac{\partial}{\partial y}$ and y momentum equation $\partial/\partial x$

$$\frac{\partial}{\partial t}\left(\frac{\partial u}{\partial y}\right) + \frac{\partial}{\partial y}(\vec{V}\nabla\vec{u}) = -\frac{1}{\rho}\frac{\partial p}{\partial x}\left(\frac{\partial}{\partial y}\right) + \frac{\nu}{\rho}\left(\frac{\partial}{\partial y}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)\right)$$

$$(-)$$

$$\frac{\partial}{\partial t}\left(\frac{\partial v}{\partial x}\right) + \frac{\partial}{\partial x}(\vec{V}\nabla v) = -\frac{1}{\rho}\frac{\partial p}{\partial y}\left(\frac{\partial}{\partial x}\right) + \frac{\nu}{\rho}\left(\frac{\partial}{\partial x}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)\right)$$

$$\frac{\partial}{\partial t}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) + \nabla\cdot\nabla(\omega) = \frac{1}{Re}\left(\nabla^2\omega\right)$$

$$\frac{\partial \omega}{\partial t} + u\left(\frac{\partial \omega}{\partial x}\right) + v\frac{\partial \omega}{\partial y} = \frac{1}{Re}\left(\frac{\partial^2\omega}{\partial x^2} + \frac{\partial^2\omega}{\partial y^2}\right)$$

$$\frac{\partial \omega}{\partial t} + \left[\frac{\partial \varphi}{\partial y}\left(\frac{\partial \omega}{\partial x}\right) + \frac{\partial \varphi}{\partial x}\frac{\partial \omega}{\partial y}\right] = \frac{1}{Re}\left(\frac{\partial^2\omega}{\partial x^2} + \frac{\partial^2\omega}{\partial y^2}\right).$$

$$\frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} = -\omega$$

$$\left[\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2}\right] = \rho\left[\left(\frac{\partial^2\varphi}{\partial x^2}\right)\left(\frac{\partial^2\varphi}{\partial y^2}\right) - \left(\frac{\partial^2\varphi}{\partial x\partial y}\right)^2\right]$$

$$Re = \frac{UL}{\nu}$$

Discretize equations

Stream Function Formula.

$$u = \frac{\partial \psi}{\partial y} \Rightarrow \frac{\psi_{i,j+1} - \psi_{i,j-1}}{2\Delta y}$$

$$v = \frac{\partial \psi}{\partial x} \Rightarrow \frac{\psi_{i+1,j} - \psi_{i,j}}{2\Delta x}$$

Vorticity Transport equation

$$\frac{\partial \omega}{\partial t} + \left[u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} \right] = \frac{1}{Re} \left[\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right]$$

① Convection term :- (central difference)

$$\left[u \frac{\partial \omega}{\partial x} + v \frac{\partial \omega}{\partial y} \right] = \left[\frac{u^n_{i+1,j} \omega^n_{i+1,j} - u^n_{i-1} \omega^n_{i-1,j}}{2\Delta x} \right] + \left[\frac{v^n_{i,j+1} \omega^n_{i,j+1} - v_{i,j+1} \omega^n_{i,j+1}}{2\Delta y} \right]$$

② Diffusion term. (central difference)

$$\frac{1}{Re} \left[\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right] \Rightarrow \frac{1}{Re} \left[\frac{\omega^n_{i+1,j} - 2\omega^n_{ij} + \omega^n_{i-1,j}}{(\Delta x)^2} + \frac{\omega^n_{i,j+1} - 2\omega^n_{ij} + \omega^n_{i,j-1}}{(\Delta y)^2} \right]$$

For the temporal solution, Explicit Scheme.

$$\frac{w_{ij}^{n+1} - w_{ij}^{n}}{\Delta t} \Rightarrow \frac{\partial w_{ij}}{\partial t}$$

Substituting the Transport equation,

$$\frac{w_{ij}^{n+1} - w_{ij}^{n}}{\Delta t} + \frac{u_{i+1j}^{n} w_{i+1j}^{n} - u_{i-1j}^{n} w_{i-1j}^{n}}{2\Delta x} + \frac{v_{ij+1}^{n} w_{ij+1}^{n} - v_{ij-1} w_{ij-1}}{2\Delta y}$$

$$= \frac{1}{Re}\left[\left[\frac{w_{i+1j}^{n} - 2w_{ij}^{n} + w_{i-1j}^{n}}{\Delta x^2}\right] + \left(\frac{w_{ij+1}^{n} - 2w_{ij} + w_{ij-1}}{\Delta y^2}\right)\right]$$

$$w_{ij}^{n+1} = w_{ij}^{n} + \frac{\Delta t}{Re}\left[\frac{w_{i+1j}^{n} - 2w_{ij}^{n} + w_{i-1j}^{n}}{\Delta x^2} + \frac{w_{ij+1} - 2w_{ij} + w_{ij-1}}{\Delta y^2}\right]$$

$$- \frac{\Delta t}{2\Delta x}\left[u_{i+1j}^{n} w_{i+1j}^{n} - u_{i-1j}^{n} w_{i-1j}^{n}\right] - \Delta t\left(\frac{v_{ij+1}^{n} w_{ij+1}^{n} - v_{ij-1}^{n} w_{ij-1}^{n}}{2\Delta y}\right)$$

For the Poisson's equation

$$\frac{\psi_{i+1}^{n} - 2\psi_{ij}^{n} + \psi_{i-1j}^{n}}{(\Delta x)^2} + \frac{\psi_{ij-1}^{n} - 2\psi_{ij}^{n} + \psi_{ij-1}^{n}}{\Delta y^2} = -w_{ij}^{n}.$$

For von-neumann. stability analysis of Transport equation. [Refered From, Lid driven cavity flow using stencil based Numerical method - Hannes Juujärvi, Isak Kinnunen.

$$\Delta t \leq \frac{h^2 Re}{4}$$

$h = \Delta x = \Delta y$. for uniform grid.